

Notation & Prerequisites

- ethose-exe

This section covers the essential notation and background math required for the course. We are skipping lengthy derivations to focus on the tools you will actually use in code.

1.1 Essential Notation

Establishing a consistent mathematical language is the first step toward understanding machine learning papers.

Scalars

A scalar is a single real number. In this course, scalars are denoted by lowercase letters like a , b , or x . If you see $x \in \mathbb{R}$, it simply means x is a member of the set of all real numbers.

Absolute Value

The absolute value $|a|$ represents the distance of a number from zero, effectively removing any negative sign.

$$|a| = \begin{cases} a & \text{if } a \geq 0 \\ -a & \text{if } a < 0 \end{cases}$$

Intervals

When variables exist within a specific range, we use brackets to define the boundaries:

- $[0, 1]$ includes both 0 and 1 (Closed interval).
- $(0, 1)$ excludes 0 and 1 (Open interval).
- $[0, 1)$ includes 0 but excludes 1 (Half-open interval).

1.2 Trigonometry Refresher

Trigonometry is fundamental in machine learning for calculating similarity and understanding data orientation.

Core Functions

- $\cos(\theta) = \frac{\text{Adjacent}}{\text{Hypotenuse}}$
- $\sin(\theta) = \frac{\text{Opposite}}{\text{Hypotenuse}}$
- $\tan(\theta) = \frac{\text{Opposite}}{\text{Adjacent}}$

These functions define relationships between the angles and sides of a right triangle, which are the basis for analyzing vector components.

Figure 1.1: Trigonometric Functions in the Unit Circle. For point (x, y) on the unit circle ($R = 1$).

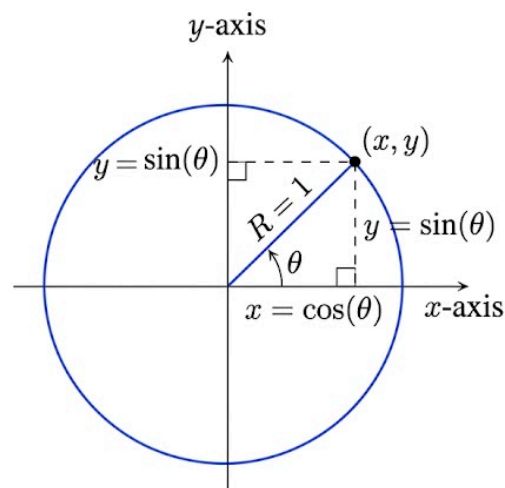


Figure 1.1: Trigonometric Functions in the Unit Circle. For point (x, y) on the unit circle ($R = 1$), $x = \cos(\theta)$ and $y = \sin(\theta)$.

Figure 1.1: Trigonometric Functions in the Unit Circle. In a unit circle (radius 1), the coordinates of any point (x, y) are $(\cos \theta, \sin \theta)$. This visual relationship is the foundation for understanding how vectors decompose into their components.

The Trigonometric Unity

The following identity holds true for any angle:

$$\sin^2(\theta) + \cos^2(\theta) = 1$$

Law of Cosines

When dealing with triangles that do not have a right angle, the Law of Cosines is used to find missing side lengths or angles. This is the mathematical basis for calculating distances between vectors.

$$c^2 = a^2 + b^2 - 2ab \cos(\gamma)$$

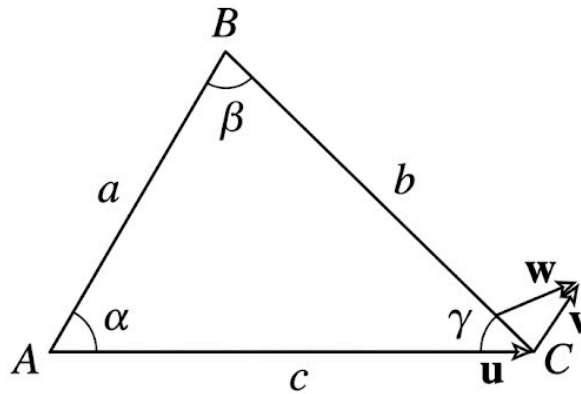


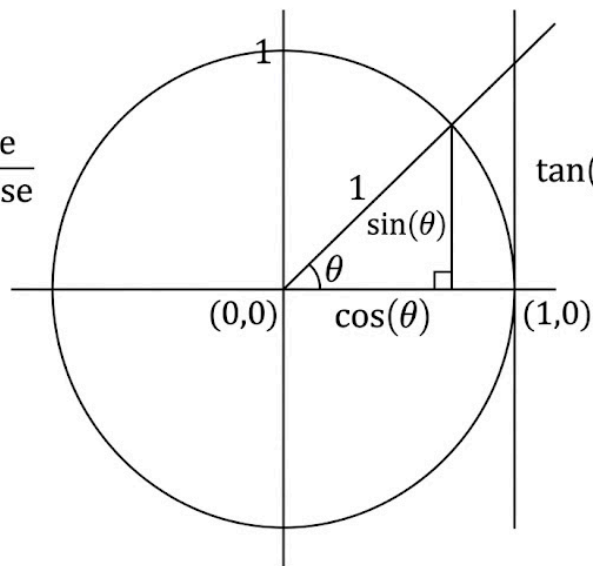
Figure 1.2: General Triangle for the Law of Cosines. In this arbitrary triangle, $c^2 = a^2 + b^2 - 2ab \cos \gamma$. When $\gamma \neq 90^\circ$, this is necessary to compute the distance between two vectors.

Figure 1.2: The Law of Cosines Geometry. This diagram shows a general triangle. When an angle (γ) is not 90 degrees, the Law of Cosines relates the lengths of the three sides (a, b, c). This is crucial for computing the distance between two arbitrary data vectors in a high-dimensional space.

Inverse Functions

Inverse functions allow you to find an angle from a ratio. For example, if $\cos(\theta) = x$, then $\theta = \arccos(x)$. In programming, these functions typically return a value within a specific range, such as $[0, \pi]$ for $\arccos$.

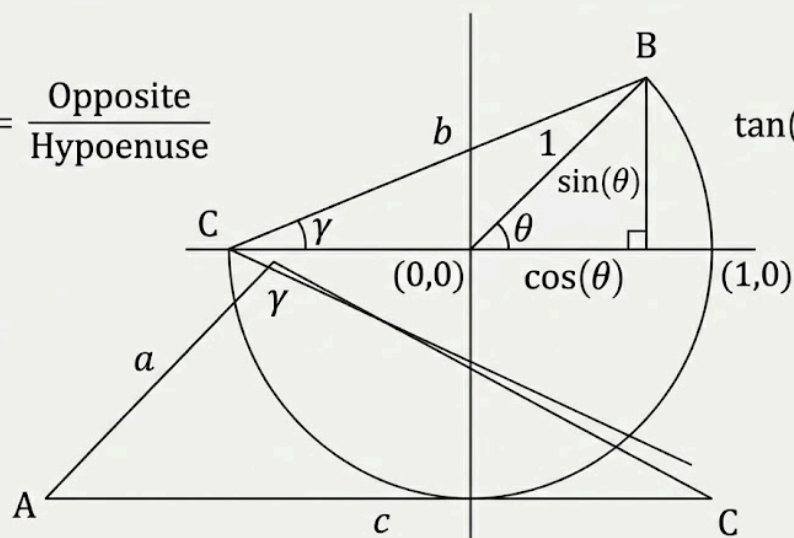
$$\sin(\theta) = \frac{\text{Opposite}}{\text{Hypothenuse}}$$



$$\tan(\theta) = \frac{\text{Opposite}}{\text{Adjacent}}$$

$$c^2 = a^2 + b^2 - 2ab \cdot \cos(\gamma)$$

$$\sin(\theta) = \frac{\text{Opposite}}{\text{Hypothenuse}}$$



$$\tan(\theta) = \frac{\text{Opposite}}{\text{Adjacent}}$$

Python Implementation

In machine learning, we use libraries like NumPy to handle these calculations. Note that most libraries expect angles in radians rather than degrees.

```
import numpy as np

# Convert 45 degrees to radians
angle_rad = np.radians(45)

# Calculate sine and cosine
s = np.sin(angle_rad)
c = np.cos(angle_rad)

# Verify identity:  $\sin^2 + \cos^2 = 1$ 
print(s**2 + c**2)
```

(Day 2 will move from these individual values into Vector Spaces and Linear Transformations.)